MERAJ MAMMADOV

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EDUCATION

PhD in Computer Science Aug 2025 – Present
Örebro University, Sweden

MSc in Mechanical Engineering 2025
Ulsan National Institute of Science and Technology (UNIST), South Korea
GPA: 97.50/100 · *summa cum laude*

BSc in Mechanical and Aerospace Engineering 2023
Ulsan National Institute of Science and Technology (UNIST), South Korea
GPA: 93.30/100 · *magna cum laude*
Multidisciplinary Major in Computer Science and Engineering

PUBLICATIONS

RAIL: Representation-Aligned Imitation Learning for Student-Compatible Teacher Policies 2026
M. Mammadov, P. Zuidberg Dos Martires, and J. A. Stork.
Advances in Neural Information Processing Systems (NeurIPS).

Teacher–Student Representational Alignment for Reinforcement Learning-Driven Imitation Learning 2026
M. Mammadov, P. Zuidberg Dos Martires, and J. A. Stork.
ICRA Workshop on Reinforcement Learning for Imitation Learning (RL4IL). **Oral presentation**.

PACMAN: Rapid Identification of Keypoint Patch-Based Fiducial Markers in Occluded Environments 2025
T. Park, G. Bae, W. Shin, **M. Mammadov**, J. Seo, H. Shin, and H. Oh.
Image and Vision Computing.

Autonomous Landing on a Moving Platform Using Vision-Based Deep Reinforcement Learning 2024
P. Ladosz, **M. Mammadov**, H. Shin, W. Shin, and H. Oh.
IEEE Robotics and Automation Letters (RA-L). **Presented at IROS**.

End-to-End LiDAR-Driven Reinforcement Learning for Autonomous Racing 2024
M. Mammadov and H. Oh.
Korea Robotics Conference (KRoC).

EXPERIENCE

Doctoral Researcher — Örebro University Aug 2025 – Present
Research: Knowledge transfer across agents with different capabilities

- Investigating how knowledge from foundation models can accelerate policy learning on new tasks
- Developed RAIL, a transfer learning framework that enables student agents to imitate better-informed teachers

Master's Researcher — Autonomous Systems Lab, UNIST Sep 2023 – Jun 2025
Thesis: Learning End-to-End Perception-Aware Policies for UAV Collision Avoidance in Dynamic Environments

- Developed an autoregressive imitation learning method with quantile networks to capture diverse teacher behavior
- Built an Isaac Sim training pipeline supporting parallel training of 2,048 agents
- Achieved 82% navigation success in dynamic environments, versus under 10% for standard RL from vision

Research Assistant — Autonomous Systems Lab, UNIST Mar 2022 – Jun 2023
Project: Vision-based autonomous landing of UAVs on moving platforms using reinforcement learning

- Developed a flight simulator in PyBullet for training RL policies to land on moving platforms using camera images
- Built Crazyflie hardware setup, tuned flight controllers, and deployed RL policies in real-world flight experiments
- Implemented conventional landing controllers and benchmarked RL policies in simulation and real-world flight tests

Summer Intern — BP, Rig Engineering Team Jul 2022 – Sep 2022
Project: Evaluation of computer vision systems for worker safety on oil rigs

- Compared four systems on detection accuracy, coverage, reliability, integration requirements, and cost
- Recommended a camera-based worker detection system to management, leading to its adoption on oil rigs in Azerbaijan
- Led vendor discussions and inspected offshore sites to define deployment requirements and assess operational risks

Research Assistant — Innovative Thermal Engineering Lab, UNIST Jun 2021 – Jan 2022
Project: Dynamic modeling and efficiency analysis of organic Rankine cycles integrated with hydrogen fuel cells

- Built a library in Modelica from scratch for transient modeling of pumps, turbines, and heat exchangers
- Developed a MATLAB tool for visualizing transient system dynamics, adopted by other researchers in the lab
- Benchmarked the efficiency of fuel cells against conventional boiler-based cycles

HONORS & AWARDS

WASP Graduate Fellow, Wallenberg AI, Autonomous Systems and Software Program, Sweden	2025 – Present
Pathways@RSS Fellow, Robotics: Science and Systems (RSS), TU Delft, Netherlands	2024
Global UniStar Scholarship for academic excellence, UNIST, South Korea	2019 – 2023
Bronze Medal, 50th International Physics Olympiad (IPhO), Israel	2019
Gold Medal, National Physics Olympiad, Azerbaijan	2019
Third Place, 3rd International Metropolises Olympiad, Russia	2018
Member of Azerbaijan's national team of five, 49th International Physics Olympiad (IPhO), Portugal	2018
Silver Medal, National Physics Olympiad, Azerbaijan	2018

REVIEWING SERVICE

International Conference on Learning Representations (ICLR)	2027
Conference on Robot Learning (CoRL)	2026
IEEE/ASME Transactions on Mechatronics	2026
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	2026
IEEE International Conference on Robotics and Automation (ICRA)	2026

TEACHING EXPERIENCE

Programming Fundamentals, Örebro University	Fall 2026
Discrete Mathematics, UNIST	Spring 2024
Introduction to Artificial Intelligence Programming II, UNIST	Fall 2023
Introduction to Artificial Intelligence Programming I, UNIST	Spring 2023

SKILLS

Tools	PyTorch, JAX, ROS, Isaac Sim, SolidWorks
Programming languages	Python, C++, MATLAB, Modelica
Spoken languages	English (C2), Azerbaijani (Native), Turkish (Native), German (A2)
Misc	Magic tricks, lockpicking, origami, chess